LECTURE 2.3A · SHIP AND OCEAN SYSTEMS DESIGN

Main Dimensions and Form Coefficients

What each main dimension and form coefficient does to the ship and to its cost, before we sweep them, with the container ship view · Papanikolaou (2026) §2.2–2.12 · Session W2.3 · Lecture about 85 min + Exercise 2.3 50 min

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From One Ship to a Domain of Ships

LECTURE 1.5 · ONE POINT

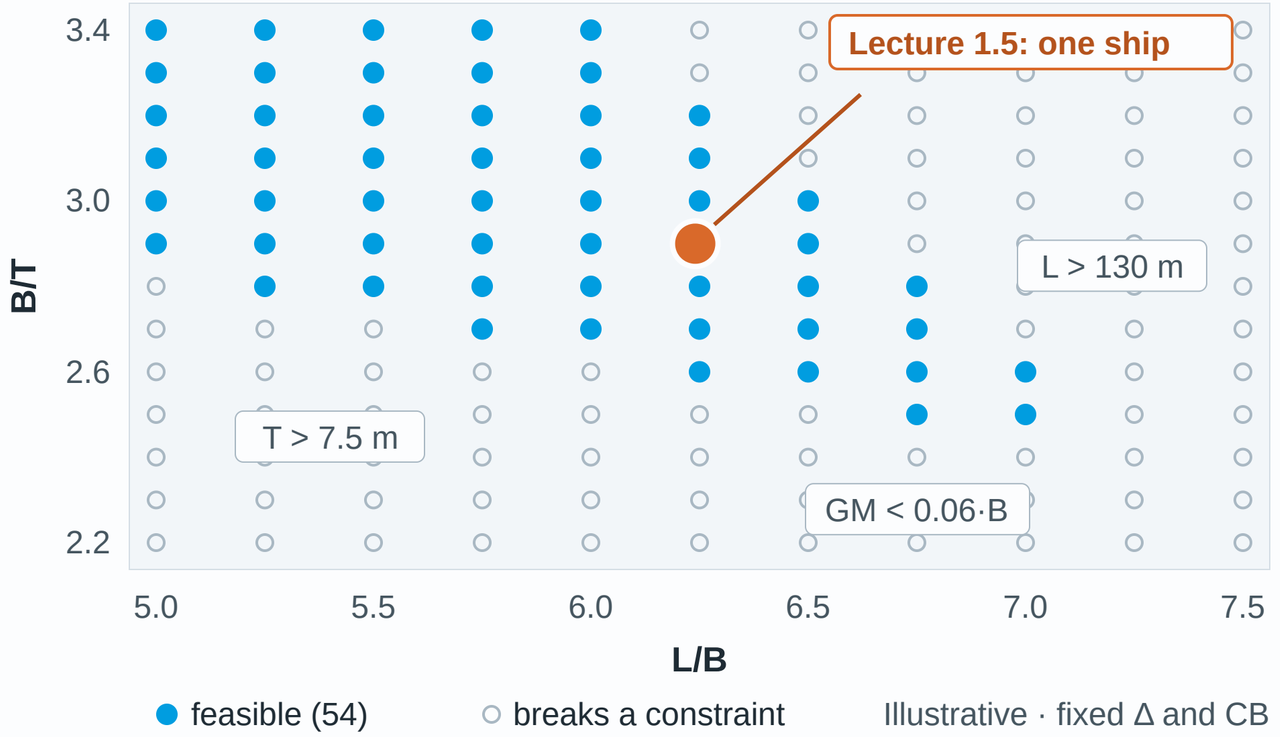
Fixed L/B, B/T and CB: the design equation solved by iteration. One ship.

NEXT · A GRID

The same equation solved for every pair in a discrete range of ratios: a domain of feasible solutions.

LATER · AN OBJECTIVE

Every feasible ship scored by an objective function, such as the least required freight rate.



Today, before the sweep: what each ratio does to the ship and to its cost; then the form coefficients.

Before the Sweep: What Each Parameter Does

1	2	3	4
Length	**Beam**	**Depth and draught**	**Form coefficients**
Friction against waves, the hull girder and its steel, slenderness, four ways to a first length	The product B·T, the stability lever, resistance and steel, and the roll it brings	The cheapest dimension; the propeller, the ballast condition and the ports	CB, CP, CM and CWP: what each changes, and how to estimate them

Every chapter ends with the container ship view: bays, rows and tiers set the dimensions, and fine hulls must carry square boxes.

Read: Papanikolaou (2026) §2.2–2.12; Tables 2.2–2.17; Fig. 2.27

Bring: a laptop and the reference ship

Every Choice Is Judged on Five Criteria

1 Hydrodynamics

Resistance and propulsion, seakeeping, manoeuvring

2 Stability

Intact and damaged, at small and large angles

3 Hold volume

Enough space for the cargo, in whole units if unitized

4 Strength

Bending, torsion, local loads: steel weight

5 Building cost

Steel, outfit, machinery and man-hours

Who judges the cost

The yard: least building cost.

The owner: building and operating cost together: least Required Freight Rate, highest NPV or ROI.

What mainly sets each dimension

L displacement and speed · **CB** the Froude number

B stability · **D** hold volume, freeboard · **T** propeller, freeboard, ports

§2.2: wrong main dimensions can hardly be corrected later. The grid lets us explore them while they are still cheap to change.

Length: the Expensive Dimension

A longer ship, same Δ	What happens	Cost	§
Resistance	Wave resistance falls, friction rises	Fuel −	2.3.1
Strength	Bending moment grows with L: thicker plates	Steel +	2.3.2
Outfit	Longer pipes and cables, more anchor gear, a larger rudder	Outfit +	2.3.3
Machinery	Less power at speed	Engine −, fuel −	2.3.4
Spaces	Number and length of holds; whole cargo units	Capacity	2.3.5
Seakeeping	Heave and pitch resonate at wave length 1.0–1.3 L	Speed loss	2.3.6

$$Fn = V / \sqrt{(g \cdot L)}$$

Length sets the Froude number, so the speed that matters is relative, not absolute (§2.3.1, conclusion 1).

The trade: length buys fuel and pays in steel. The book: *the smallest ship that meets the requirements is generally the cheapest to build* (footnote 13).

Length Against Resistance: Friction or Waves?

RT = RF + RR · RF = ½·CF·ρ·S·V², CF from the ITTC 1957 line

Lengthen by 10 %, same Δ and V

Friction grows with the wetted surface: **RF × 1.03**

Residuary falls with Fn: **RR × 0.87** (exponent α = 4; range 3–5)

For Fn ≥ 0.15 the residuary drop wins: RT falls (Eqs. 2.10–2.14)

Reference ship, Fn 0.21, RF ≈ 65 % of RT

RT × (0.65 · 1.03 + 0.35 · 0.87) = **0.97**: 3 % less resistance

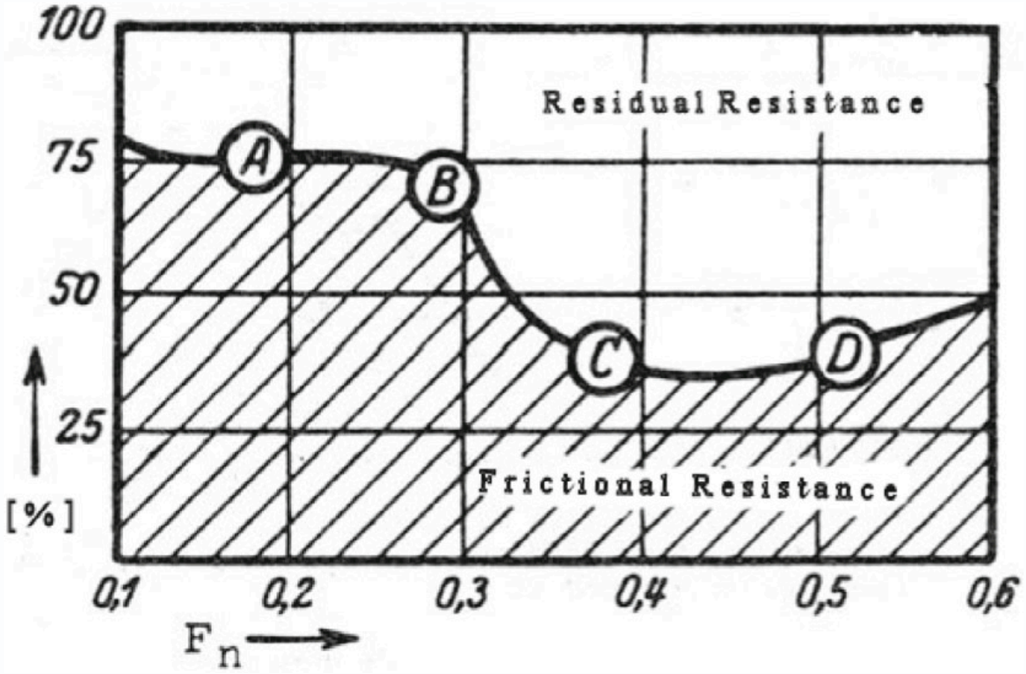


Fig. 2.9 · Share of friction and residuary resistance against Fn (Horn 1930). A slow cargo ship, B liner, C small passenger ship, D torpedo boat.

Slow ships (Fn ≤ 0.20): friction rules, so short and full. Fast ships (Fn ≥ 0.25): waves rule, so long and fine (§2.3.1).

Bow and Stern Waves Decide the Good Speeds

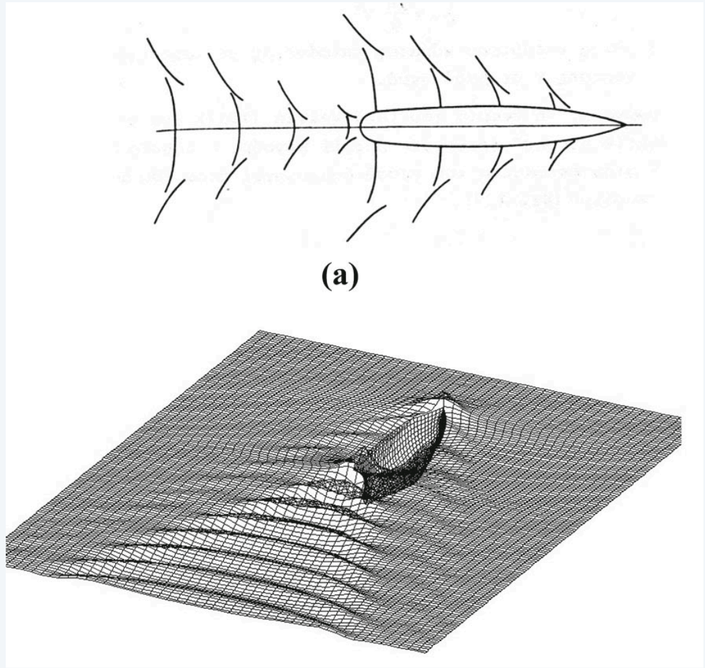


Fig. 2.12 · (a) bow and stern systems; (b) a container ship, panel method

$$\lambda_w = \frac{2\pi V^2}{g} \quad (2.15) \qquad \frac{L_{WL}}{\frac{1}{2}\lambda_w} = \frac{1}{\pi F_n^2} = 2n+1, \quad n = 1, 2, 3, \ldots \quad (2.17)$$

Odd: bow and stern waves cancel, favourable · **even (2n):** they add, adverse

λw wave length [m] · V speed [m/s] · g = 9.81 m/s² · LWL waterline length [m] · Fn Froude number

Fn	RF / RT	CW curve	Typical ship (Table 2.3)
0.15	80 %	hump	VLCC
0.19	70 %	hollow	medium tanker
0.23	60 %	hump	medium-speed cargo
0.25	60 %	hollow	fast cargo
0.29–0.31	50 %	hump	fishing vessel
0.33–0.36	40 %	hollow	fast cargo, reefer

Past the critical speed power grows faster than V³. Service speed stays below it; trial speed ≈ 1.06·VS needs ≈ 1.25·P (2.21).

A bad Fn? Change the length, smooth the shoulders, or change the speed (§2.3.1).

Length Costs Steel

M ≈ C·Δ·L (2.23) · cargo ships: C ≈ 0.012 in calm water, + 0.025 hogging in waves

Lengthen by λ, same Δ	Plates	Hull steel	+10 % L
A · B, T and midship section fixed, CB falls by 1/λ	t × λ	× λ³⁄²	+15 %
B · CB fixed, B·T falls by 1/λ	t × λ²	power 2.5	+27 %

Same 10 % as slide 6: **about −3 % resistance for +15 to +27 % hull steel.** Fuel is paid every year, steel once: the objective function weighs them.

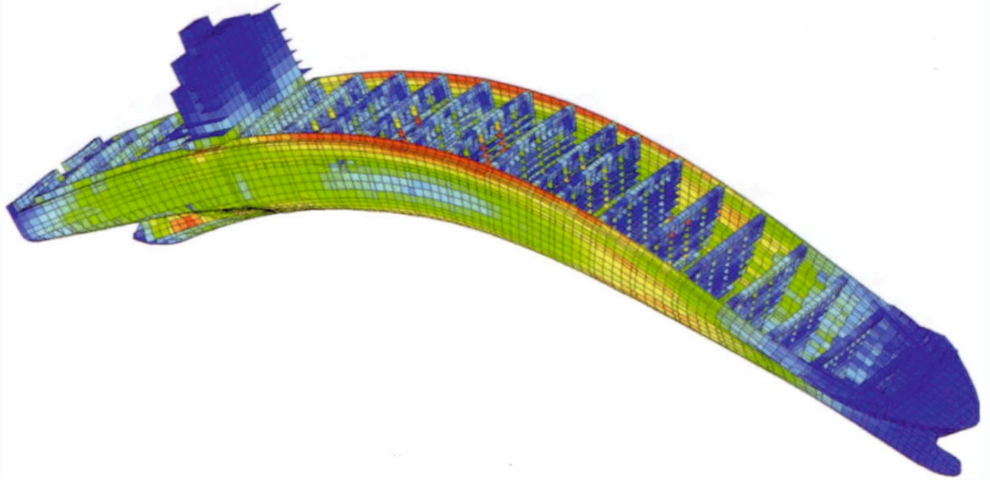


Fig. 2.18 · A container ship hogging, FEM (Germanischer Lloyd)

Wave moments grow with L²·B (IACS UR S11, 2.24–2.25). Open-deck ships also twist, and torsion grows with length (Fig. 2.20).

Shifting displacement from length into beam or draught helps the girder, but moves the problem to transverse strength (§2.3.2).

Slenderness and the Ratios of Similar Ships

$L/\nabla^{1/3}$ with CP describes the hull

High $L/\nabla^{1/3}$ and low CP: a fine, fast hull, small waves (§2.4.1).

Low $L/\nabla^{1/3}$ and high CB: least wetted surface, simple and cheap structure, but watch transverse strength (§2.4.2).

Only both together make a slender hull: liner 7.2 with CP 0.57; tug 5.2 with 0.62; river cargo ship 6.8 with 0.85 (§2.10).

Ship type (Table 2.4)	$L/\nabla^{1/3}$	L/B	B/T	CB
Fast seagoing cargo	5.6–5.9	5.7–7.8	2.2–2.6	0.56–0.64
Slow seagoing cargo	5.2–5.4	4.8–8.5	2.1–2.3	0.65–0.73
Bulk carriers	4.7–5.6	5.0–7.1	2.1–3.2	0.72–0.86
Tankers, Fn 0.16–0.18	4.5–5.2	5.0–6.5	2.2–2.9	0.78–0.86
LNG carriers	5.6–5.9	6.0–6.6	2.4–4.0	0.73–0.80
Fast reefers	6.1–6.5	6.7–7.2	2.8–3.0	0.57–0.59
Ferries	6.2–6.9	5.9–6.2	3.7–4.0	0.50–0.60

Ranges are one standard deviation about the regression (App. A). Container ships: Fig. 2.27, slide 11.

These ranges are where a grid of ratios starts: around similar ships, then wider where the objective may pay off.

Four Ways to a First Length

A Slenderness $L/\nabla^{1/3}$ of similar ships (Table 2.4, App. A)

B Least building cost: $L = C \cdot (\Delta \cdot V)$ to the power 0.3, then +10% for running cost (Schneekluth, 2.45)

C Fleet statistics: Ayre (2.46), Posdunine–van Lammeren (2.47), Völker (2.48)

D Diagrams by ship type: TEU (Fig. 2.27), hold volume (Fig. 2.26)

Reference ship, 14 kn	L [m]
A · $L/\nabla^{1/3}$ 5.2–5.4	122–127
B · Schneekluth	125.8
B · plus 10%	138.4
C · Ayre	126.7
C · Posdunine, cargo (LWL)	128.5
C · Völker	132.1
Exercise 1.5 result	**124.8**

Δ 13,200 t, $\nabla^{1/3}$ = 23.44 m. The old formulas run long (footnote 15).

Container Ships: Length Comes in Bays

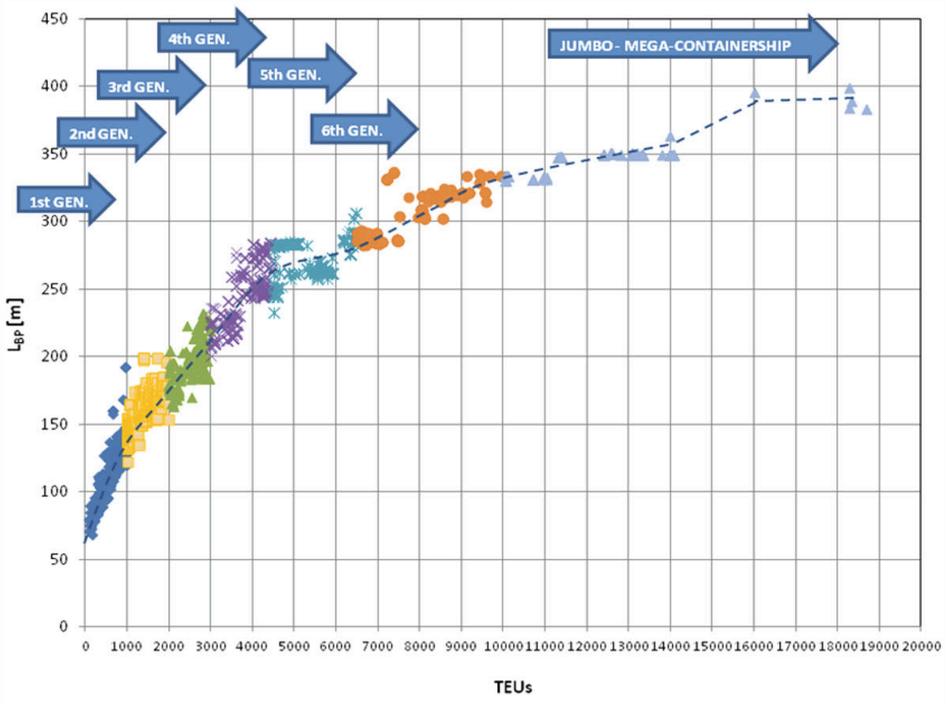


Fig. 2.27a · LPP against TEU, by generation (Papanikolaou 2014)

Capacity first, then length

Bays of 40 ft fix the hold length; add the engine room and the peaks. Little freedom at the ends (§2.3.5).

Fast cargo ships: check Fn

596 TEU feeder, L ≈ 113 m: Fn 0.22 at 14 kn · 0.25 at 16 · 0.28 at 18 · **0.31 at 20 kn**, on the 0.29–0.31 hump.

Length twists

Open decks: torsion grows with L (Fig. 2.20). New Panamax: L < 366 m.

Designed as fast ships, often operated slower (footnote 32): check the length at both speeds.

Beam and Draught Share One Product

$B \cdot T = \nabla / (L \cdot CB)$ (2.49) · Reference ship: 137.8 m² = 20.0 m × 6.89 m, L/B 6.24, B/T 2.90

Choose L/B …

From similar ships (Table 2.4) or Friis et al.: cargo 4 + 0.015·(L + 17); container ships 4 + 0.009·(L + 42) for 100–200 m, 6.5–7.1 above (2.56, 2.58)

Lower L/B turns better, keeps course worse (§2.6.4)

… or B/T

About 2.5: least wetted surface, and it serves wave resistance too (§2.5)

Above 2.5 when the draught is limited (large tankers, bulk carriers) or stability demands it (RoPax); twin screw above 2.6

Well below 2.5 is rare

… then solve two equations

$B/T = L^3 \cdot CB / ((L/B)^2 \cdot \nabla)$ (2.51)

$B \cdot T = \nabla / (L \cdot CB)$ (2.52)

Two equations, two unknowns: B and T

Beam and draught are not independent choices: more of one means less of the other, until a route limit or the freeboard breaks the tie.

Beam Is the Stability Lever

$$BM = IT / \nabla \cdot IT \approx kT \cdot L \cdot B^3,\ kT\ 0.04–0.06 \quad (2.69–2.70)$$

$$\delta BM / BM = 3 \cdot \delta B / B \rightarrow \delta GM \approx 3 \cdot BM \cdot \delta B / B \quad (2.54–2.55)$$

Reference ship, BM 4.59 m

Beam +5 % (draught adjusted, Δ fixed): **GM + 0.69 m, from 2.3 to about 3.0 m**

KB and KG do not depend directly on B (2.66–2.68). If T falls instead of CB, KB drops a little and KG falls too: GM still rises.

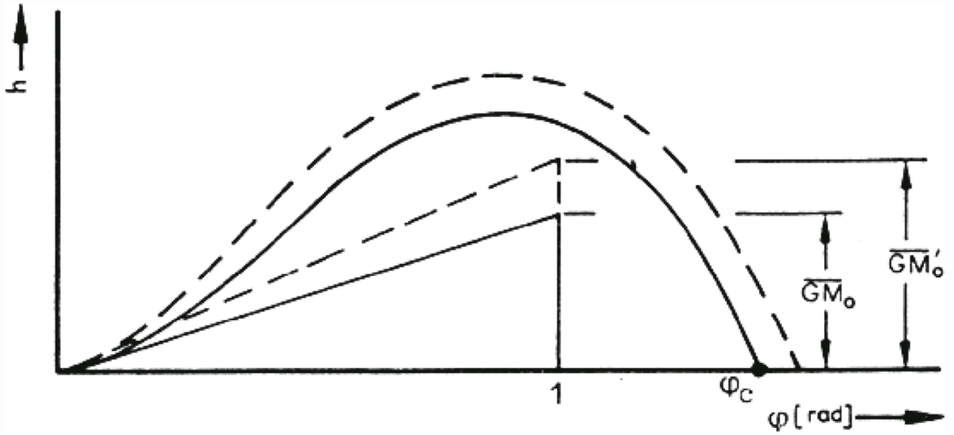


Fig. 2.36 · More beam: larger GM, and usually a larger GZmax and range

Large angles

Wider at the same section area, the deck edge immerses earlier, yet GZmax and the range usually grow (Fig. 2.35).

Cost: beam is the cheapest stability there is; §2.12.4 prefers it to a fuller waterplane. Its price is roll (slide 15) and, slightly, steel (slide 14).

Beam, Resistance and Steel: Less Simple Than It Looks

Resistance (§2.6.2)

Fast ships: more beam, steeper waterlines, more wave resistance.

Slow ships: least wetted surface at B/T ≈ 2.5 and CB ≈ 0.80.

Mumford: RT changes with (B1/B0) and (T1/T0) to powers set by Fn and ship type (2.71, Tables 2.8–2.9).

The "paradox": a 22 kn reefer, Fn 0.315

Same L and T	L/B 6.5	7.0	7.5
CP	0.56	0.60	0.65
RT / RT0	0.80	1.00	1.32

Wider at the same L and Δ means a finer CP, and less resistance. Longer and finer helps more (Table 2.11: 0.79).

Steel (§2.6.3)

Wider at the same D: thinner plates, but more shell: usually slightly more steel.

Wider and shallower: L/D up, thicker plates, more steel. Weight concentrated amidships raises the bending moment.

Other factors (§2.6.4)

Unit cargo sets the beam in steps; canals cap it; a draught limit pushes it up.

Lower L/B: tighter turning, worse course keeping.

Too Much Stability Also Hurts

$T\varphi = 2\pi \cdot i\varphi / \sqrt{(g \cdot GM)}$, $i\varphi \approx 0.38 \cdot B$ (2.74) · Kempf: $R = 2\pi \cdot i\varphi / \sqrt{(B \cdot GM)}$, good between 8 and 14 (2.75)

More beam, more GM, shorter roll period

$T\varphi$ falls like $1/\sqrt{B}$. Short periods tune with short, frequent waves and bring large transverse accelerations high up: on the deck stacks of a container ship (§2.6.4).

Reference ship, GM 2.3 m

$T\varphi \approx 10$ s, $R \approx 7.0$: stiff. With $i\varphi = 0.38 \cdot B$, R from 8 to 14 means **GM between about 0.03·B and 0.09·B**.

Natural roll period (Table 2.13)	s
Cargo ships	12–18
Coastal cargo ships	7–10
Bulk carriers	12–20
Tankers	≈ 20
Reefers	16–18
RoPax ferries	10–14
Open-sea tugs	8–12

Parametric roll in head and following seas: GM swings as the waterplane changes in the waves, worst with large flare. Container ships are especially sensitive (APL China 1998, Maersk Essen 2021).

Container Ships: Beam Comes in Rows

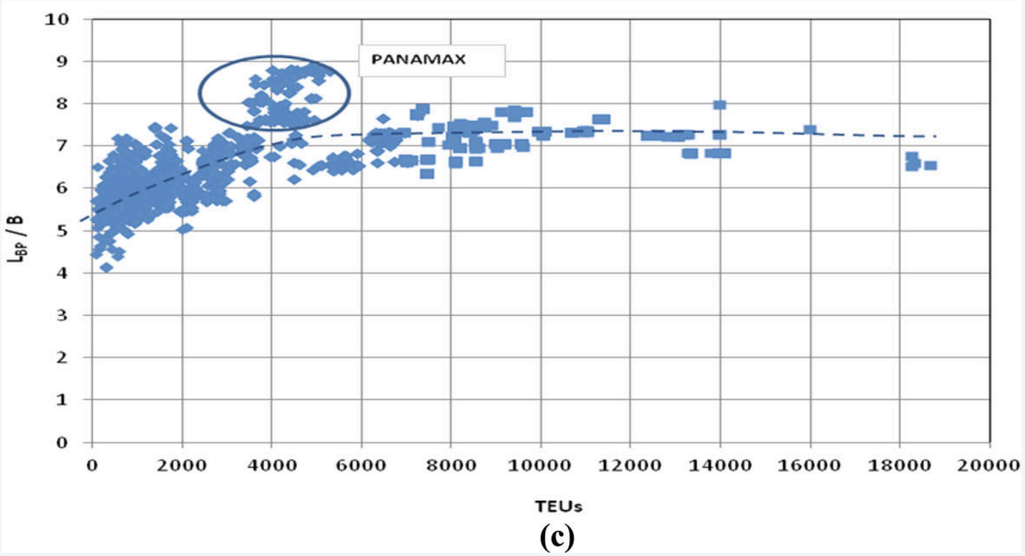


Fig. 2.27c · L/B against TEU. Circled: the Panamax ships, beam capped at 32.31 m, run to L/B 7.5–9.

Old Panamax B < 32.31 m · New Panamax B < 49 m · Ever Given, 400 × 59 m, blocked Suez in 2021.

Rows across (2.50)

B ≈ 3·n + 2.2 m, n rows in the hold, near old Panamax size.

596 TEU feeder: 6 rows → 20.2 m; real 20.0 m.

KMTC Seoul, 32.5 m: 2.50 says 10 rows; counting rows of 2.44 m between 2.15 m wing tanks gives 11 (Lecture 3.2).

Boxes on deck

KG is high, so beam buys the GM; too much GM gives violent roll and lashing loads (slide 15).

> Beam moves in steps of one row, about 2.5 m: a grid of B/T should step with it.

Depth: the Cheapest Dimension

+8%

hold volume for +10% depth at L/D = 10; +4% at L/D = 14, for a small gain in steel

What sets D

Hold volume: for containers, the tiers of 8′6″ plus double bottom and coaming

Freeboard: D − T at least the Load Line minimum

Class limit L/D about 14–16, beyond that a special strength check

What D does

KG rises, so GM falls; the range of stability grows

Section modulus grows, so plates can be thinner

Displacement hardly changes

Where the rules come in (§2.7.1): Load Line freeboard, the SOLAS bulkhead deck, tonnage and the class L/D limit all act on depth. Lecture 2.3b, to be scheduled.

Draught: Propeller, Ballast and Ports

Resistance and propulsion (§2.8.1)

B/T 2.0–2.5 for least wetted surface; about 2.5 serves wave resistance too.

Deeper: a larger, slower propeller, higher efficiency; but it breaks the surface more easily in ballast. Twin screw: B/T above 2.6.

Stability (§2.8.2)

Deeper at the same Δ: KB rises; with V sections and a fuller waterplane BM rises too.

The same hull loaded deeper: IT grows less than ∇, so BM falls.

Ballast, seakeeping, manoeuvring (§2.8.3)

Forward TF ≥ 0.02·L against slamming; aft TA ≥ DP + e + 0.4 m against racing (2.76–2.77).

Rudder: L·T / AR = 30–80 (2.78).

Strength (§2.8.4)

Deeper: higher bottom pressure, more bottom steel; small if L can shrink.

Deeper and wider: transverse strength.

Routes (§2.8.5): Panama 12.04 m old, 15.2 m new · Suez 18 m · Kiel 9.5 m · St Lawrence 7.6 m · La Plata 8.2 m. A draught limit pushes the beam up.

Container Ships: Depth Comes in Tiers

D + coaming ≥ double bottom + n · box height · 8′6″ = 2.59 m, high cube 9′6″ = 2.90 m (§2.7.2)

596 TEU feeder

0.8 m + 4 high cubes = 12.38 m ≤ D 11.3 m + coaming 1.6 m = 12.9 m

L/D = 10.0: the boxes set D, far from the class limit of 14–16.

On deck

Deck tiers are not set by D: stability, lashing and the bridge view (SOLAS V/22) set them (Lecture 3.2).

Draught

Volume carriers: boxes weigh less than their maximum, so the ship often sails below its scantling draught. Design for both.

Canal and port depths decide the class: old Panamax 12.04 m, New Panamax 15.2 m.

Depth is the cheap dimension, but for containers it moves in whole tiers of 2.6–2.9 m.

Five Numbers Describe the Hull Form

Coefficient	Definition	What it tells
CB	∇ / (L·B·T)	Fullness against the box
CP	∇ / (AM·L) = CB / CM	Where the volume sits: low CP, volume amidships and fine ends
CM	AM / (B·T)	Fullness of the midship section: 0.94–0.99 for cargo ships
CWP	AWP / (L·B)	Fullness of the waterplane: stability and wave making
L/∇⅓	slenderness	With CP: how slender the hull really is

Also: LCB, KB, the parallel body LP, the entrance and run LE and LR; and qualitatively the section shapes, bow, stern, freeboard and sheer (§2.2).

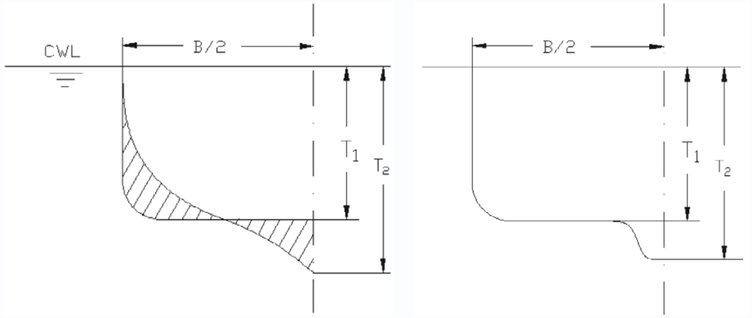


Fig. 2.44 · Same CP and Δ, deeper and narrower sections: CM and CB fall a lot. On tugs and fishing boats CB alone misleads.

§2.9: CP describes the form better than CB, so it should be chosen first; CM, CB and CWP follow.

CP Follows the Froude Number

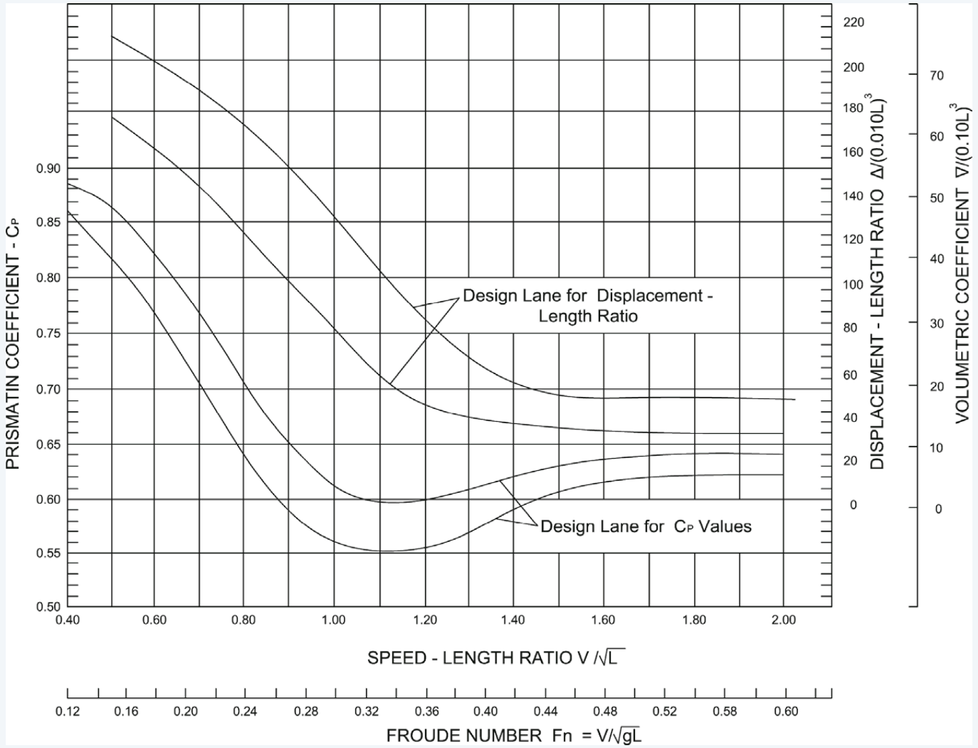


Fig. 2.45 · CP and volumetric coefficient of built ships against Fn (Saunders)

Fn	How CP is chosen (§2.10.1)
≤ 0.24	Above the optimum: friction rules; cost and space decide
0.24–0.36	Near the hydrodynamic optimum: fuel weighs as much as building cost
0.36–0.46	Resistance hardly depends on CP
> 0.46	Resistance falls slowly as CP grows: lift and trim decide

Why fine at speed

Displacement concentrated amidships softens the shoulders and their wave systems (§2.3.1).

Slow ships

Least wetted surface: CP and CB up to 0.88, CM up to 0.99 (tankers, bulk carriers).

Estimating CB, CM and CWP

CB = K1 − K2·Fn − K3·Fn² (2.86)	K1	K2	K3
Horn · single screw, service speed	1.06	1.68	0
Ayre · single screw, trial	1.08	1.68	0
Ayre · twin screw, trial	1.09	1.68	0
Heckser · single screw, trial	1.00	1.44	0
van Lammeren · single screw, trial	1.08	1.68	0.224

Least building cost (Schneekluth, 2.88): CB = (0.14 / Fn) · (L/B + 20) / 26, for 0.14 ≤ Fn ≤ 0.32

Gives	Formula
CM	0.9 + 0.1·CB (van Lammeren, 2.90); Kerlen, HSVA (Table 2.14)
CM	1 − rB² / (2.33·B·T), bilge radius rB (2.94)
CWP, U sections	0.778·CB + 0.248 (2.81)
CWP, normal	(1 + 2·CB) / 3 (2.83) · 0.743·CB + 0.297 (2.84)
CWP, V sections	higher; 0.793·CB + 0.297 (§2.12.3)

Service speed ≈ 0.94 × trial speed. CWP formulas are for cruiser sterns: a transom gives more, corrected by K (2.100).

CB Beyond Resistance

Seakeeping (§2.10.2)

Head seas, λ ≈ 1.05·L, same power: speed loss **22 % at CB 0.5, 55 % at CB 0.7** (Todd).

Small CP: more green water; needs freeboard, bow height and flare. Added resistance can reach 70 % of calm water.

Building cost (§2.10.3)

Slender ships need more steel and outfit per cubic metre of hold: high CB and CP are cheaper to build.

Spaces (§2.10.4)

Holds fill best with flat bottoms and vertical sides; small CB with V sections wastes the ends.

Containers: cells stepped at the ends (slide 26).

Stability (§2.10.5)

Small CP with larger B and T and V sections raises KB and IT, so GM.

Practice: CB above the hydrodynamic optimum, for cost and space (§2.10). How far above is a question for the objective function.

CM: Small Effect, Many Trade-offs

A fuller midship section	Effect	§
Resistance	Small and indirect, through CP: more wetted surface and bilge eddies. Slow ships: least wetted surface near CM 0.80, but draught limits push CM up.	2.11.1
Building cost	Lower: flat panels, a constant bilge radius, less bending	2.11.2
Space	More, for break bulk and unit cargo	2.11.3
Stability	Lower: small CM with V sections, wider or deeper, raises stability	2.11.4
Roll damping	More: square sections damp roll; small CM needs bilge keels	2.11.5

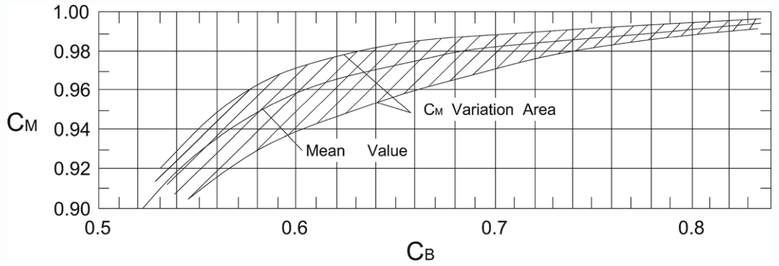


Fig. 2.56 · CM against CB

Bilge keels: width ≥ 2 % of B, length about 25 % of L; aligned with the flow by tests or CFD.

Cargo ships: bilge radius 2.0–2.7 m (Table 2.17).

Practice: CM follows CB and falls for small CB and high Fn (Fig. 2.56). Fast ships may even use bulbous sections, CM > 1.

CWP: Stability Now, Resistance Later

$$IT = CIT \cdot L \cdot B^3 / 12 \cdot CIT = CWP \cdot (0.1272 + 0.8724 \cdot CWP) \quad (2.95) \cdot GM \geq \text{about } 0.06 \cdot B$$

Reference ship

CWP 0.83 → CIT 0.71; CWP 0.80 → 0.66: **BM −7 %**, the same as 2.3 % less beam.

Resistance and seakeeping (§2.12.2)

Slow ships: full waterlines forward, local CWP 0.90–0.95; finer aft, about 0.80, for the propeller flow.

Fast ships: fine ends; the optimum CWP falls as Fn rises. Full bow waterlines slam in head seas.

§2.12.4, three warnings

1 · Buy stability with beam, not with CWP.

2 · Do not assume a high CWP early: it shrinks when the lines are drawn, and so does GM.

3 · A transom brings a high CWP, lost when the stern emerges in waves: GM drops, parametric roll.

Container Ships: Fine Hulls for Square Boxes

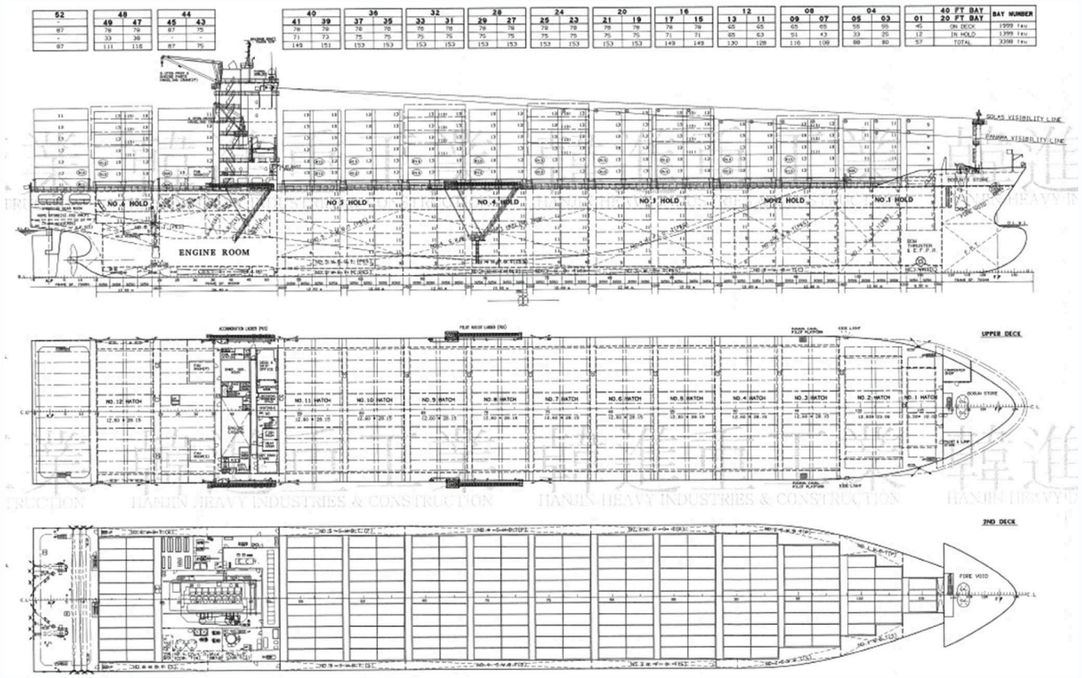


Fig. 2.47 · A 3,400 TEU ship: cells step in at the fine ends; note the SOLAS and Panama visibility lines.

A box cargo in a fine hull

Fn calls for small CP and CB; the boxes want a box. The cells follow the sloping sides at the ends (§2.10.4).

The price of fine ends

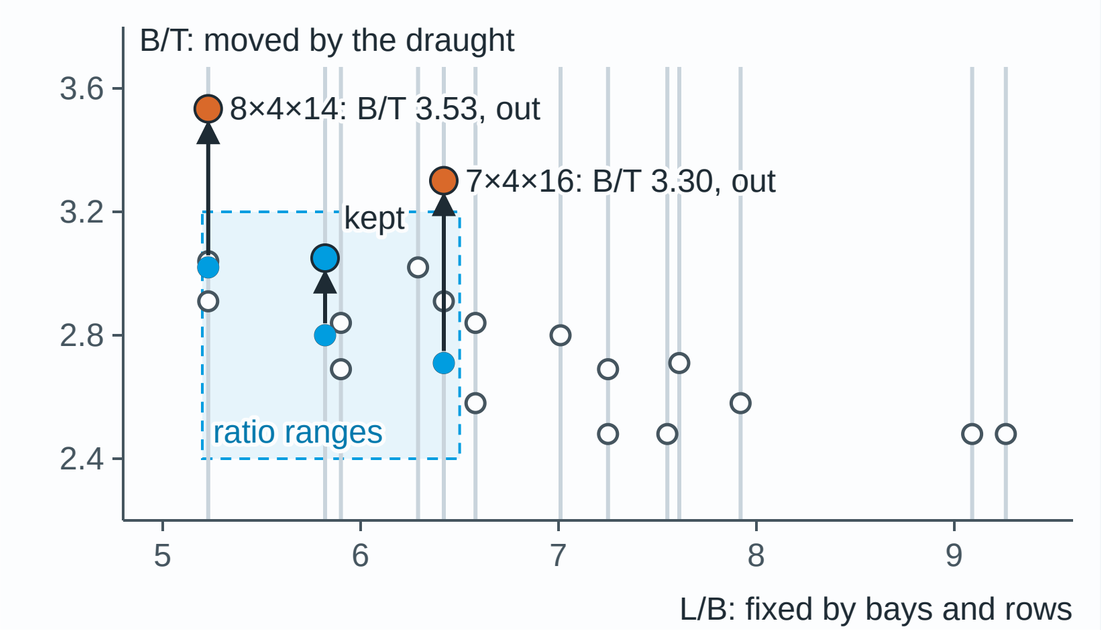
596 TEU feeder: 680 TEU in a box hull, 586 in the CB 0.68 hull: **−14 %** (Lecture 3.2).

CM and CWP

Not chosen to fit a few more boxes (§2.11.3). Flare and a wide transom carry deck rows but invite parametric roll.

> An optimization for least power, least ballast and most deck boxes gave V-shaped sections and a low CM (CONTiOPT, footnote 31).

Container Ships: a Lattice, Not a Region



The 20 candidates of a 596 TEU sweep. Open: B/T at the target draught. Arrows: the draught that floats the weights; only 7 × 4 × 14 stays in the box.

Deadweight carrier · slide 2

Any L/B and B/T on the grid: the design equation scales the hull to the DWT. Every point is a ship.

Container ship

Bays, rows and tiers fix L, B and D: L/B exists only on these lines. The equation moves T and CB alone, along the line.

No small steps

A failed hull is not nudged: its neighbour is one row, one tier or two bays away, ≈ 2.5 m of B, 2.9 m of D, 13 m of L.

Enumerate the lattice, or search it with integer variables: bays, rows and tiers in a genetic algorithm (Priftis et al. 2018).

What Each Parameter Buys, and What It Costs

More …	Resistance, fuel	Stability	Steel, building cost	Space	Seakeeping
L, same Δ	▼ for Fn ≥ 0.15	–	▲ +15 to 27 % per +10 %	▲ holds, bays	resonance λ ≈ L
B, B·T fixed	fast ▲, slow ≈ at B/T 2.5	▲▲ GM + 3·BM·δB/B	▲ slightly	▲ rows	▲ roll accelerations
T, B·T fixed	▲ propeller efficiency	BM ▼	▲ bottom	–	ports, ballast racing
D	–	GM ▼, range ▲	▼ thinner plates; cheapest	▲ +8 % per +10 %	freeboard ▲
CB, CP	slow: friction ▼ · fast: waves ▲	▼ (small CP helps)	▼ cheaper	▲	speed loss ▲
CM	▲ a little	▼	▼ cheaper	▲ (not for boxes)	roll damping ▲
CWP	▲ wave making	▲ BM	–	▲ deck area	slamming ▲

Blue: gain · orange: cost · arrows show the direction of the quantity. The objective function puts a price on every cell.

Checking the Reference Ship

Check	Value	From the book	Verdict
Length	124.8 m	122–138 m, the four ways	Short end of the cluster
Froude number	0.206	≤ 0.24: slow ship	CB above optimum is normal
CB	0.749	0.70–0.73 (Heckser, Horn, van Lammeren); 0.69 Schneekluth	Fuller: say why
CM · CP	0.985 · 0.76	CM 0.975–0.992 (Table 2.14)	In range
CWP	0.83	0.833, normal sections	In range
L/B	6.24	6.13, Friis cargo formula	In range
B/T	2.90	about 2.5 for least resistance	Wider: why?
L/D	12.9	at most 14–16 (class)	In range
$L/\nabla^{1/3}$	5.32	5.2–5.4 slow cargo (Table 2.4)	In range
GM, first look	2.3 m	at least 0.06B = 1.2 m (§2.12.1)	In range

The Story in Four Lines, and a Lens

1 · Length. Buys fuel, pays in steel: −3 % resistance for +15–27 % hull steel per +10 %. Choose by Fn.

2 · Beam. B·T is fixed; beam is the stability lever (3·BM·δB/B), with roll as its price.

3 · Depth and draught. Depth is cheap volume; draught buys a better propeller until a port says no.

4 · Coefficients. CP from Fn; CB above the optimum for cost and space; stability from beam, not CWP.

Container ships: bays, rows and tiers set L, B and D in steps; the fine hull costs slots; deck boxes make roll and parametric roll design issues.

Next: these effects become the axes and constraints of the grid. Then Exercise 2.3 (slide 31). Lecture 2.3b, Law and Regulations, is to be scheduled.

EXERCISE 2.3 · INDIVIDUAL · 50 MIN · PRACTICE FOR D3

Main Dimensions of the Reference Ship, Checked

The reference ship, the 10,000 t general cargo ship: Δ 13,200.7 t, 124.8 × 20.0 × 6.89 m, D 9.64 m, 14 kn. Ayre: $L/\nabla^{1/3} = 3.33 + 1.67 \cdot V/\sqrt{L}$ (V in kn, L in m, iterate). Völker: $L/\nabla^{1/3} = 3.5 + 4.5 \cdot V/\sqrt{(g \cdot \nabla^{1/3})}$ (V in m/s). Your own file; keep slides 10, 22 and 29 closed until step 4.

1 Length · 10 min

From Δ and 14 kn: L from slenderness 5.2–5.4, Schneekluth and +10%, and Ayre or Völker (formulas above). Where does 124.8 m sit?

2 Coefficients · 10 min

Fn on LPP; CB by Horn, Heckser and Schneekluth (2.88); CM 0.985, CP and CWP for normal sections (slide 22).

3 New input: 15 kn · 10 min

The owner asks for 15 kn. Keep L 124.8 m, B 20.0 m and Δ: new Fn, CB by Horn, T = ∇/(L·B·CB). Does T clear the 7.5 m port limit?

4 Check table · 10 min

Fn, L/B, B/T, L/D, $L/\nabla^{1/3}$ and CB for both versions against the book (slide 29). Which values need a reason?

Check: the worked answer is shown at the end (Exercise Answers – Week 2). Not graded.
Your project: your group applies this to its own ship outside class and completes D3 (see your Project Scope Sheet).